

International Islamic University Chittagong
DBMS Midterm Lab Test Spring 2025
Instructor : Muhammad Mubinur Rahman (MUR)
Course Name: Database Management Systems Lab
Course Code: CSE-2424
Time: 1 hour
Section : 4BM

DBMS Midterm Lab Question

Total Marks: 20

This question set is aimed at solving Complex Engineering Problem. In this question set of 20 marks, we are aiming at achieving:

Course Learning Outcome (CLO) 1- Demonstrate familiarity with SQL, PL/SQL, NoSQL and fundamentals of Database Management System.

Program Outcome (PO) 1- Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.

Knowledge Profile K3- Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

Complex Engineering Problem Attributes P1, P3 and P4:

P1 - Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3-K6 or K8 which allows a fundamentals-based, first principles analytical approach PO1- PO7.

P3 - Depth of analysis required- Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.

P4 - Familiarity of issues: Involve infrequently encountered issues.

Questions are presented in the next page.

Question 1 (7 Marks)

Create two tables for a **Gym Management System**:

- **Members:** `member_id` (Primary Key), `first_name`, `last_name`, `membership_fee`, `branch_id` (can be NULL).
- **Branches:** `branch_id` (Primary Key), `branch_name`.

Insert at least **5 members** and **3 branches**.

Make sure at least **one member has no branch assigned**.

Also insert one branch that has no members.

Question 2 (6 Marks)

Write a query to:

- Display each member's **average membership fee rounded up** to the nearest whole number.
 - Show the **membership fee increased by 10%**, rounded to the nearest whole number.
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Question 3 (7 Marks)

Write a query to:

- Display each member's **full name** and their **branch name**.
 - Do not Include members who do not belong to any branch.
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